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Workshop No. 1 — Project Definition and Source Research

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Public Opinion and Sentiment on Climate Change and Environmental Policies

1. Topic Angle Selection

1.1. Objective

The objective of this project is to analyze how people express their opinions and emotions regarding climate change, environmental policies, and ecological events through digital platforms. The goal is not only to determine whether opinions are positive or negative, but also to identify patterns in how people react, discuss, and perceive these topics in different contexts.

To achieve this, data will be collected from sources such as online forums and news media, where individuals freely share their perspectives. Using this data, natural language processing techniques will be applied to interpret textual content and extract relevant information about sentiment and dominant ideas.

This analysis will provide a clearer understanding of society's level of environmental awareness as well as the most common concerns related to climate change. Additionally, it can support organizations and researchers in better understanding public opinion and making informed decisions based on real data.

1.2. Topic Angle Definition

The focus of this project is to analyze public opinion on climate change from two different perspectives that allow us to understand how environmental perception is formed and communicated. On one hand, discussions on Reddit will be analyzed. Reddit is an online platform where people share content, express opinions, and participate in discussions on a wide range of topics. It works as a large forum divided into communities called subreddits (for example r/climatechange) , where each community focuses on a specific subject. This makes it possible to capture real reactions, emotions, and debates among users about environmental issues (Hugging Face datasets will be used as a complementary resource to validate results. However, they are not part of the main pipeline.). On the other hand, articles from international news media will be analyzed, which present information in a more structured and informative way.

The project focuses on comparing these two sources to identify differences between public opinion and the way media outlets present climate change. This approach allows for a more complete understanding of the topic and helps explain how social perception around environmental issues is shaped.

1.3. Topic Justification

Climate change is one of the most pressing global issues today, as it directly impacts the environment, the economy, and people's quality of life. In the public sphere, this topic generates a wide range of opinions and sentiments, spanning from deep concern and a sense of urgency for action to skepticism and misinformation.

Furthermore, discussions regarding environmental policies and ecological events often spark intense debates among different groups, reflecting diverse stances on responsibility, potential solutions, and the consequences of climate change. These opinions are constantly expressed on digital platforms and in the media, making this topic a rich source of data for analyzing social perceptions, emotions, and trends.

1.4. Data Language and NLP Implications

The primary language of the data will be English, as the selected subreddit (r/climatechange) and international news sources publish content predominantly in this language. This provides access to a larger volume of information and more diverse discussions regarding climate change.

From a Natural Language Processing (NLP) perspective, working with English data offers a significant advantage, as there are more tools, pre-trained models, and resources available for this language. This facilitates tasks such as sentiment analysis, text classification, and topic modeling.

However, it also involves addressing certain challenges, such as the presence of informal language, abbreviations, sarcasm, or grammatical errors on platforms like Reddit, which can impact the precision of the analysis.

1.5. Time Scope of the Analysis

The topic of climate change is considered ongoing, as it is a global issue that is constantly discussed across various public spheres. Unlike other topics that rely on specific occurrences, conversations regarding climate change, environmental policies, and ecological events remain active over time.

However, it is important to note that specific events—such as natural disasters, environmental summits, or new policy announcements—can trigger spikes in public opinion and reactions. These moments can significantly influence both the intensity and the tone of the discussions.

2. API Source Research: Reddit

2.1. API Identification

For this project the Reddit API will be used. This is a REST API that provides access to posts, comments, and metadata from various online communities. Specifically, we will work with the r/climatechange subreddit, which contains active discussions related to climate change and environmental issues.

2.2. API Documentation

Base URL: `https://www.reddit.com/`

Authentication Method: Public access (no authentication required for basic requests)

Available Endpoints:

- `/r/climatechange.json` → To retrieve recent posts.
- `/r/climatechange/top.json` → To retrieve the most popular posts.
- → `The project focuses on retrieving post data such as titles and metadata from the subreddit.`

Rate Limits: Reddit limits the number of requests to prevent overhead, especially when full authentication is not used.

Data Format: JSON

Available Fields:
- `title` (Post title)
- `selftext` (Post content/body)
- `author` (User who posted)
- `created_utc` (Timestamp/Date)
- `score` (Votes/Popularity)
- `num_comments` (Number of comments)

2.3. API Justification

The Reddit API is appropriate for this project because it provides access to authentic user opinions on climate change, allowing for the direct analysis of public sentiment and perception.

The retrieved data includes raw text, such as posts and comments, which are ideal for applying Natural Language Processing (NLP) techniques. Furthermore, metadata such as dates, authors, and interaction counts allows us to enrich the analysis and observe social trends over time.

2.4. Accessibility Verification

A request was made to the Reddit API using the endpoint corresponding to the `r/climatechange` subreddit, resulting in a successful response in **JSON** format. This confirms that the API is accessible and allows for the retrieval of information relevant to our analysis.

Below is a snippet of the obtained response:


```
{
 "subreddit": "climatechange",
 "title": "Learning about climate change, how problematic is it?",
 "author": "Degenerate2Throwaway",
 "score": 9,
 "num_comments": 40,
 "created_utc": 1774128948,
 "selftext": "I'm starting to look into how bad things are or aren't..."
}
```


Additionally, other examples of relevant posts were identified within the same response:

- *"Sweden's 'old-growth' natural forests store 83 % more carbon than managed woodlands—new study"*
- *Records shattered as summer heat hits Southwest in March"*
- *"So could we just replace oil/coal/Natural Gas as an energy source"*

These data points include both textual content and metadata (author, date, number of comments, and score), making them highly useful for the sentiment analysis and public opinion research tasks required for this project.

3. Web Scraping Source Research

3.1. Site Identification

The .Environment"section of **The Guardian** (https://www.theguardian.com/environment) was selected as the secondary data source. This website is a publicly accessible platform that provides high-quality articles, expert analysis, and in-depth reporting on the climate crisis. It offers structured textual content that serves as a professional contrast to social media discussions.

3.2. Ethics and Legality Verification (robots.txt ToS)

- robots.txt: The file was verified at `theguardian.com/robots.txt`. The site allows generic crawlers to index its news and environment sections, provided that administrative and internal search areas are not accessed.
- Terms of Service: The Guardian permits the use of its content for personal and non-commercial academic purposes. To ensure an ethical and sustainable process, the following principles are applied: (*rate limiting*) to avoid overloading their servers.

3.3. Ethical Considerations and Technical Limitations

To ensure an ethical and sustainable data collection process the following principles have been integrated into the pipeline design:

- API Compliance and Rate Limiting: The ingestion scripts include sleep timers and retry logic to strictly adhere to Reddit's and The Guardian's developer terms of service, preventing any potential Denial of Service (DoS) impact on their platforms.
- Data Privacy: Only publicly available data is persisted. No Personal Identifiable Information (PII) beyond usernames or public author names is stored, ensuring compliance with general data protection best practices.
- Language Bias: We acknowledge that the current pipeline is limited to English-speaking sources. While this covers a significant portion of the global climate discourse, it may exclude perspectives from non-English speaking regions, which is a known limitation for future iterations.
- Data Traceability: By persisting the original URL for every record, we ensure that every sentiment score can be traced back to its source, preventing the creation of "black-boxresults.

3.4. Target Pages and HTML Structure

The extraction will focus on individual articles within the `/environment` category. The following HTML structure and fields have been identified for extraction:

Title: `<h1>` tag.

Body Text: `<p>` tags located within the main article container.

Date: `<time>` tag with the `datetime` attribute.

Author: Tags containing the `rel=.ᵃuthor"` attribute.

3.5. Complementary Source Justification

This source is the ideal complement to the Reddit API due to the clear difference in discourse types:

1. Professional Narrative vs. Informal Voice: While Reddit captures the informal voice of users (spontaneous reactions, colloquial language, and emotional debates), The Guardian provides the perspective of experts, scientific data, and structured editorial discourse.
2. Trend Validation: It allows us to contrast whether topics presented in the news media generate a symmetrical impact on the perception and sentiment of digital communities.
3. Data Diversity for NLP: By combining formal news structure with natural, informal forum language, the data pipeline will be much more robust for training text classification models and identifying linguistic patterns.

4. Extraction Strategy Definition

4.1. Ingestion Periodicity

For this project, a batch ingestion strategy is implemented. Data is collected by manually executing ingestion scripts for both Reddit and *The Guardian* sources.

Justification: This approach provides flexibility during development and testing while still allowing the system to capture recent discussions and news articles related to climate change. Additionally, it avoids exceeding API limits and reduces the risk of overloading external servers. This process can be extended to automated scheduling in future iterations.

4.2. Raw Data Format and Nomenclature

All collected data is stored in **JSON** format within the Bronze layer. This format was chosen due to its flexibility in handling semi-structured textual data.

`[source]_YYYYMMDD_HHMMSS.json`

Examples:

- `reddit_20260322_101530.json`
- `news_20260322_101845.json`

This naming convention ensures traceability and prevents overwriting previously ingested data.

4.3. Data Processing and Transformation (Silver Layer)

The transformation process is implemented using Python and the NLTK library, specifically the VADER (*Valence Aware Dictionary and sEntiment Reasoner*) sentiment analysis model.

All JSON files from the Bronze layer are loaded and processed. For each record, a unified text field is constructed by combining available attributes such as `title`, `text` (Reddit), and `content` (news articles).

A sentiment score is computed using the compound value provided by VADER, which ranges from -1 (most negative) to 1 (most positive). Based on this score, each record is classified as:

- Positive: score $\geq$ 0,05

- Negative: score $\leq -0{,}05$
- Neutral: $-0{,}05 < score < 0{,}05$

The processed data is stored in the Silver layer in **Parquet** format with the following fields:

- `text`
- `sentiment_score`
- `label`

4.4. Aggregation and Metrics (Gold Layer)

In the Gold layer, aggregated metrics are generated to support analysis and visualization.

The dataset is grouped by the sentiment label (*positive, negative, neutral*) to compute the total number of records per category. Additionally, the overall average sentiment score is calculated as an indicator of the general tone of the dataset.

The results are stored as:

- A CSV file containing sentiment counts
- A text file containing the average sentiment score

4.5. Visualization and Dashboard

An interactive dashboard is implemented using Plotly Dash to visualize the results.

The dashboard reads the processed data from the Silver layer and generates a histogram based on the `label` field, allowing users to observe the distribution of sentiments across the dataset.

The interface includes a title and a graphical component, providing a clear and intuitive representation of the analysis.

4.6. Data Quality Considerations

Basic validation is applied during processing to ensure meaningful results. Only records with non-empty text are included in the sentiment analysis.

Additionally, by combining multiple textual fields into a single input, the system ensures that all available context is considered during sentiment evaluation.

While advanced techniques such as language detection or duplicate removal are not implemented, the current approach provides a reliable baseline that can be extended in future work.

4.7. Data Dictionary: Bronze Layer Schema

To ensure consistency in the ingestion process, the following schema describes the structure of the raw data stored in the `datalake_bronze/` directory.

Field Name	Data Type	Description	Example Value
`title`	String	Title of the post or news article	`Çlimate change is accelerating"`
`text`	Text	Body content of Reddit posts (if available)	`"Discussion about climate impacts..."`
`content`	Text	Full article content extracted from news sources	`"Scientists report that..."`
`score`	Integer	Popularity score (Reddit)	`150`
`num_comments`	Integer	Number of comments (Reddit)	`45`
`date`	Integer (UTC)	Publication timestamp	`1710000000`
`url`	URL	Link to the original article (news only)	`https://www.theguardian.com/...`

Cuadro 1: Raw Data Structure in Bronze Layer

5. Data Sample and Validation

5.1. Reddit Data Sample

A sample of the data collected from Reddit is presented below. The data was retrieved using the Reddit API and includes both textual content and metadata fields:

```
{
 "title": "Learning about climate change, how problematic is it?",
 "score": 9,
 "num_comments": 40,
 "created_utc": 1774128948,
 "selftext": "I'm starting to look into how bad things are or aren't..."
}
```

5.2. News Data Sample

The following example corresponds to data extracted from The Guardian environment section using web scraping techniques:

```
{
 "title": "Climate crisis impacts intensify across regions",
 "url": "https://www.theguardian.com/environment/...",
 "content": "Scientists and policymakers warn that climate change effects are becoming more severe...
}
```

5.3. Validation

A total of approximately 100 Reddit posts and 18 news articles were successfully collected. The dataset includes structured fields such as titles, content, timestamps, and engagement metrics. This confirms that both sources provide sufficient and relevant textual data for sentiment analysis and further processing within the pipeline.

6. Functional User and User Stories

6.1. Functional User

The primary functional user of this system is an environmental researcher, journalist, or data analyst interested in understanding how climate change is perceived by the public. This user requires tools to analyze large volumes of textual data in order to identify trends, sentiments, and patterns in environmental discourse.

6.2. User Stories

- As a researcher, I want to analyze sentiment trends so that I can understand how public perception evolves over time.
- As a journalist, I want to compare social media discussions with news content so that I can identify differences in narrative and framing.
- As a data analyst, I want to process large amounts of textual data so that I can extract meaningful insights.
- As a policymaker, I want to understand public reactions so that I can design better environmental policies.
- As a user, I want an interactive dashboard so that I can explore the results in a clear and intuitive way.

7. Pipeline Architecture Overview

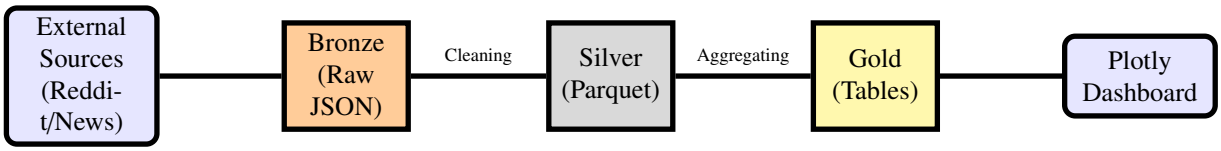

Figura 1: Data Pipeline Architecture: Medallion Strategy

Ingestion Stage: The pipeline begins by fetching raw data from the **Reddit API** and **The Guardian**. This initial step ensures we capture both spontaneous public opinions and professional journalistic perspectives, providing a balanced view of the environmental discourse.

Bronze Layer (Immutable Storage): Data is saved exactly as it arrives in **JSON** format. This layer serves as our "Historical Vault,.enSuring that the original evidence is preserved and is never lost or modified, allowing for future reprocessing if needed.

Silver Layer (The Refinement Zone): This is where the heavy lifting occurs. We apply **Natural Language Processing (NLP)** techniques to strip away "noise"(such as HTML tags, URLs, and emojis) and perform initial sentiment labeling. Data is then transformed into **Parquet** files for high-speed processing and efficiency.

Gold Layer (The Knowledge Base): Instead of storing raw text, this layer holds **aggregations and metrics**. It focuses on "The Big Picture,"such as daily sentiment averages, trend analysis, and key topic distributions, making it ready for high-level insights.

Plotly Dash (Visualization): The final destination where complex data is transformed into **easy-to-understand interactive charts**. This layer allows the end-user to explore the results of the sentiment analysis through an intuitive web interface.

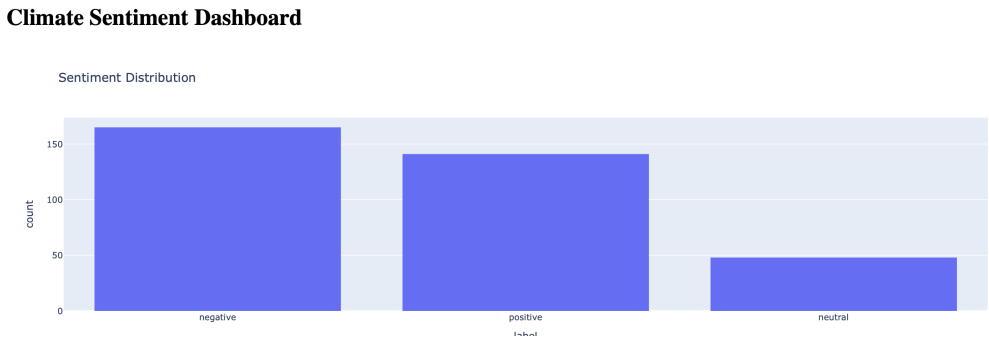

Figura 2: Sentiment distribution of climate-related posts and news articles.

8. Dashboard Design

Figure 2 illustrates the distribution of sentiment labels obtained from the processed dataset, showing the proportion of positive, negative, and neutral opinions.

8.1. Governance Dashboard

The governance dashboard is designed to monitor the quality and consistency of the data pipeline. It includes key indicators such as the number of records processed, data completeness, and general data integrity. This ensures that the system operates correctly and that the data used for analysis is reliable.

8.2. Storytelling Dashboard

The storytelling dashboard focuses on presenting insights derived from sentiment analysis. It visualizes the distribution of sentiments (positive, negative, and neutral) across the dataset and allows users to explore how public opinion is expressed in different sources.

This dashboard was implemented using Plotly Dash, providing an interactive interface for analyzing climate-related public sentiment.

Code Availability

The code used in this study is publicly available at: https://github.com/Milu2662/climate$_project$

9. Results

The sentiment analysis was conducted using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model from the NLTK library. Each text entry, constructed from titles and available content fields, was assigned a compound sentiment score ranging from -1 to 1. Based on this score, the data was classified into three categories: positive (0.05), negative (-0.05), and neutral (between -0.05 and 0.05).

The processed dataset was stored in the Silver layer in Parquet format, enabling efficient data handling and further aggregation. In the Gold layer, sentiment counts were computed by grouping records according to their assigned label, and an overall average sentiment score was calculated to summarize the general tone of the dataset.

The results show a distribution of sentiment across the three categories, reflecting differences in tone between user-generated content and news articles. Reddit data tends to exhibit more emotionally driven and polarized expressions,

while news content presents a more neutral and informative tone.

To support data exploration, the aggregated results were visualized using an interactive dashboard built with Plotly Dash. The dashboard displays a histogram of sentiment labels, allowing users to easily interpret the distribution of opinions within the dataset.

Overall, the results demonstrate that the implemented pipeline successfully transforms raw textual data into structured sentiment insights, enabling the analysis of public opinion on climate change.

10. Conclusion

This project demonstrates the implementation of an end-to-end data pipeline for sentiment analysis using real-world textual data from Reddit and online news sources. Through the integration of data ingestion, processing, and visualization stages, the system effectively converts unstructured text into meaningful analytical insights.

The use of the VADER sentiment analysis model allows for efficient classification of textual data into positive, negative, and neutral categories, providing a quantitative representation of public opinion. By storing intermediate results in Parquet format and generating aggregated metrics in the Gold layer, the pipeline ensures scalability and efficient data processing.

The results highlight observable differences between informal user-generated content and structured journalistic narratives, confirming that sentiment varies depending on the source and communication style.

Furthermore, the implementation of an interactive dashboard using Plotly Dash enhances the accessibility of the analysis, enabling users to explore sentiment distributions in a clear and intuitive way.

This approach provides a solid foundation for future extensions, such as incorporating more advanced machine learning models, expanding data sources, or applying the methodology to other domains where understanding public sentiment is relevant.

Referencias

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